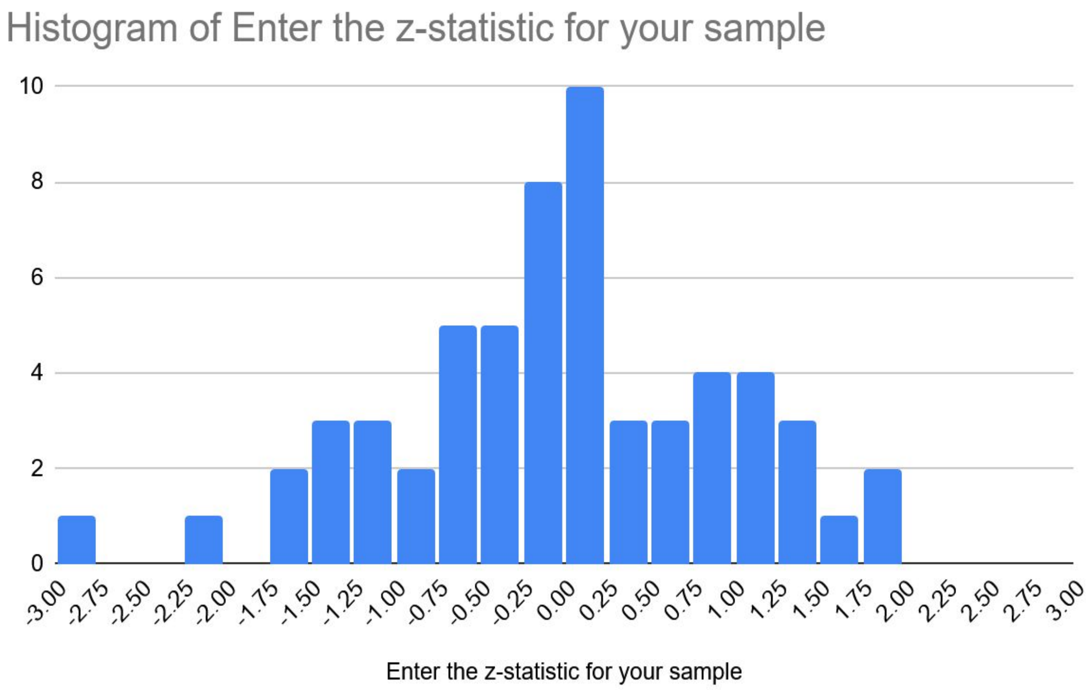
More on t - Part 1

MATH 213

Class dotplots will go here…

z-statistics

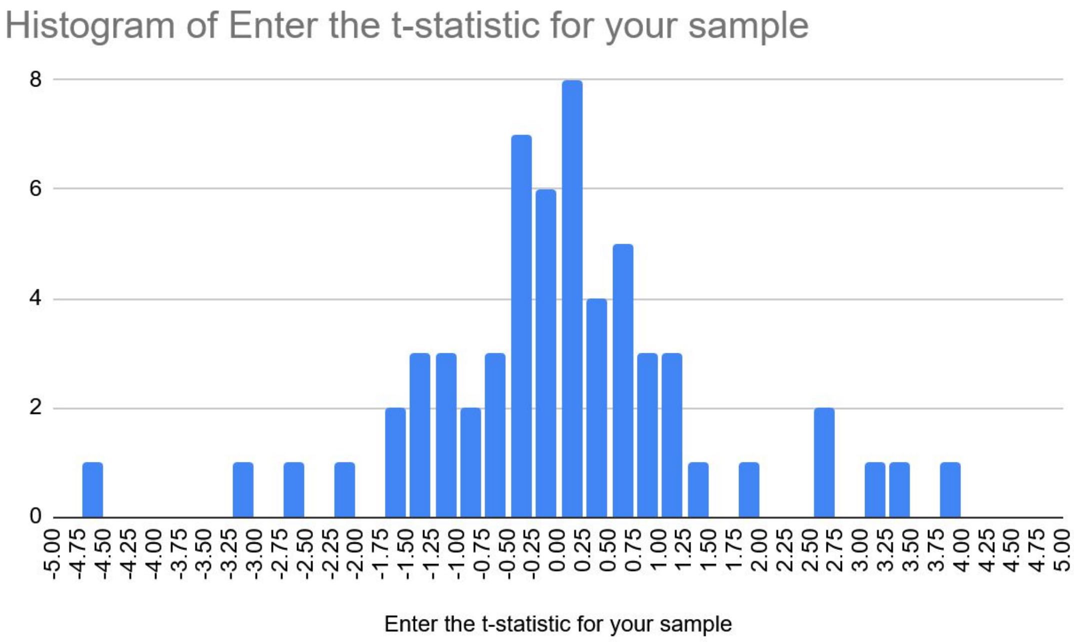
Average	-0.02658235353
Std Dev	0.9608445576
Q1	-0.60764775
Q3	0.548023325
median	0.006066969



The other class dotplot will go here…

t-statistics

Average	0.001249979383
Std Dev	1.450089504
Q1	-0.5514851
Q3	0.6908843
median	0.0028570265



Group 1

What is the difference between a t-statistic and a z-statistic? How are they calculated? What do they measure?
Both t- and z-statistics measure the number of standard errors the sample mean is from the population mean. The t-statistic is calculated with the sample standard deviation(s), the z-statistic is calculated with the population standard deviation(σ).

Look at the dotplot for our class z-statistics. What seems to be the typical size of a z-statistic for samples from the class?
0.006066969

Look at the dotplot for our class t-statistics. What seems to be the typical size of a t-statistic for samples from the class?
0.001249979383

Which type of statistic (z or t) seems to show more variability?
t

Group 2

What is the difference between a t-statistic and a z-statistic? How are they calculated? What do they measure?
The equation for z-statistic: (x_bar - mu) / (sigma / sqrt(sample_size))
The equation for t-statistic: (x_bar - mu) / (s / sqrt(sample_size))

The z-statistic uses the population standard error, so it calculates the true standard errors above the population mean that the sample mean falls, while the t-statistic uses the sample.standard error, so it's estimated instead.

Look at the dotplot for our class z-statistics. What seems to be the typical size of a z-statistic for samples from the class?
-0.027

Look at the dotplot for our class t-statistics. What seems to be the typical size of a t-statistic for samples from the class?
0.0012

Which type of statistic (z or t) seems to show more variability?
The t-statistic is more spread out (more variability) than the z-statistic.

Group 3

What is the difference between a t-statistic and a z-statistic? How are they calculated? What do they measure?

The difference is a z statistic is used when the population standard deviations known and the sample size is large. A t statistic is used when the population standard deviation is unknown and must be estimated from the sample. Both measure how far a sample mean is from the population mean in terms of standard errors, but t distribution is wider.

Look at the dotplot for our class z-statistics. What seems to be the typical size of a z-statistic for samples from the class?

Based on the dotplot, the typical size of a z-statistic seems to be around 0.006.

Look at the dotplot for our class t-statistics. What seems to be the typical size of a t-statistic for samples from the class?

Based on the dotplot, the typical size of a t-statistic seems to be roughly 0.003.

Which type of statistic (z or t) seems to show more variability?

The t statistic seems to show more variability, as the range of potential values is higher and the standard deviation is also higher.

Group 4

What is the difference between a t-statistic and a z-statistic? How are they calculated? What do they measure?

t-statistic is calculated with the population SE while z-statistic is calculated with the sample SD

They are both calculated by:(sample mean - population mean)/(SD/sqrt(sample size))

The z-statistic, or z-score, is the measure of distance the simulated sample mean is from the mean of all samples using the standard error as a measurement for distance. The t-statistic is the same but for the population's mean.

Look at the dotplot for our class z-statistics. What seems to be the typical size of a z-statistic for samples from the class?

~~-0.02658235353~~

0.006066969

Look at the dotplot for our class t-statistics. What seems to be the typical size of a t-statistic for samples from the class?

~~0.001249979383~~

0.0028570265

Which type of statistic (z or t) seems to show more variability?

T

Group 5

What is the difference between a t-statistic and a z-statistic? How are they calculated? What do they measure?

A t statistic focuses on the difference between the estimated mean and the population mean divided by the estimated standard deviation and squared sample size. Meanwhile z is calculated by doing all of that, but s is instead replaced with sigma, thus meaning that it’s the population deviation instead of the predicted standard deviation

Look at the dotplot for our class z-statistics. What seems to be the typical size of a z-statistic for samples from the class?
The 50% percentile of z values appears to be in between -0.60764775 and 0.548023325 with a center of -0.02658235353

Look at the dotplot for our class t-statistics. What seems to be the typical size of a t-statistic for samples from the class?
The 50% percentile of t values appears to be in between -0.5514851 and 0.6908843 with a center of 0.001249979383

Which type of statistic (z or t) seems to show more variability?
It’s hard to say for certain, but it seems as though t seems to have more variability due to the fact that the outliers seem to be a lot higher for the t than the z’s

Group 6

What is the difference between a t-statistic and a z-statistic? How are they calculated? What do they measure?
T-statistic: (sample mean - population mean) / (s/sqroot(n))
Z-statistic: (sample mean - population mean) / (sigma/ sqroot(n))

Look at the dotplot for our class z-statistics. What seems to be the typical size of a z-statistic for samples from the class? 0.006066969

Look at the dotplot for our class t-statistics. What seems to be the typical size of a t-statistic for samples from the class? 0.001249979383

Which type of statistic (z or t) seems to show more variability?
T statistic

Group 7

What is the difference between a t-statistic and a z-statistic? How are they calculated? What do they measure?
A z-statistic is the number of true standard errors above the population mean that your sample mean falls
- Calculated: (mean) / (sigma / sqrt(n))

A t-statistic is the number of estimated standard errors above the population mean that your sample falls
- Calculated: (mean) / (s / sqrt(n))

Look at the dotplot for our class z-statistics. What seems to be the typical size of a z-statistic for samples from the class?
0.006

Look at the dotplot for our class t-statistics. What seems to be the typical size of a t-statistic for samples from the class?
0.0012

Which type of statistic (z or t) seems to show more variability?
t-statistic

Group 8

What is the difference between a t-statistic and a z-statistic? How are they calculated? What do they measure?

The z-statistic is the true amount of standard errors that fall above the population mean, the t statistic is the estimate. The t-statistic is calculated using x_bar -mu/ s/sqrt(n), s being the standard deviation, and the z statistic is the same thing but instead of using s you use sigma.

Look at the dotplot for our class z-statistics. What seems to be the typical size of a z-statistic for samples from the class?

The average size of a z statistics is typically around the zero mark, it could be positive or it could be negative depending on whether or not the mu is bigger or smaller than your z statistic. Which the exact value for our data is 0.00606

Look at the dotplot for our class t-statistics. What seems to be the typical size of a t-statistic for samples from the class?

In between 0 and .25 is where most of the t statistics land

Which type of statistic (z or t) seems to show more variability?

The T statistic seems to show more variability with the value of it being higher than the value of the z statistic

Group 9

What is the difference between a t-statistic and a z-statistic? How are they calculated? What do they measure?

A t-statistic is used when a population's standard deviation is unknown and small, while a z-statistic is when a population’s standard deviation is known and large.

Look at the dotplot for our class z-statistics. What seems to be the typical size of a z-statistic for samples from the class?

Look at the dotplot for our class t-statistics. What seems to be the typical size of a t-statistic for samples from the class?

Which type of statistic (z or t) seems to show more variability?

Group 10

What is the difference between a t-statistic and a z-statistic? How are they calculated? What do they measure?

The z-statistic is used when the population standard deviation is known or the sample size is large, and it follows the standard normal distribution. The t-statistic is used when the population standard deviation is unknown and the sample size is small, and it follows the t-distribution with heavier tails to account for more variability.

T = (x_bar - mu) / (s / sqrt(n),   Z = (x_bar - mu) / (sigma / (sqrt(n)

Look at the dotplot for our class z-statistics. What seems to be the typical size of a z-statistic for samples from the class?

0.006066969

Look at the dotplot for our class t-statistics. What seems to be the typical size of a t-statistic for samples from the class?

Which type of statistic (z or t) seems to show more variability?